Learn Apply Grow

MODULE 8

HYPOTHESIS TESTING BASICS

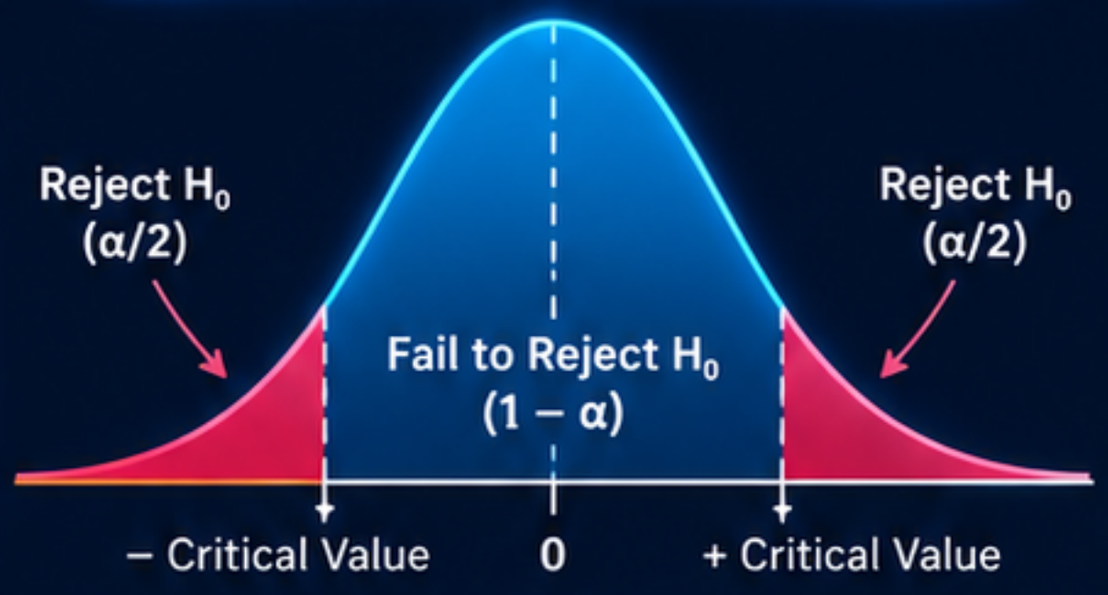
From **Assumption** to **Evidence**

> "Question assumptions,
> test with data,
> make confident decisions."
>
> - Tech With Shivansh

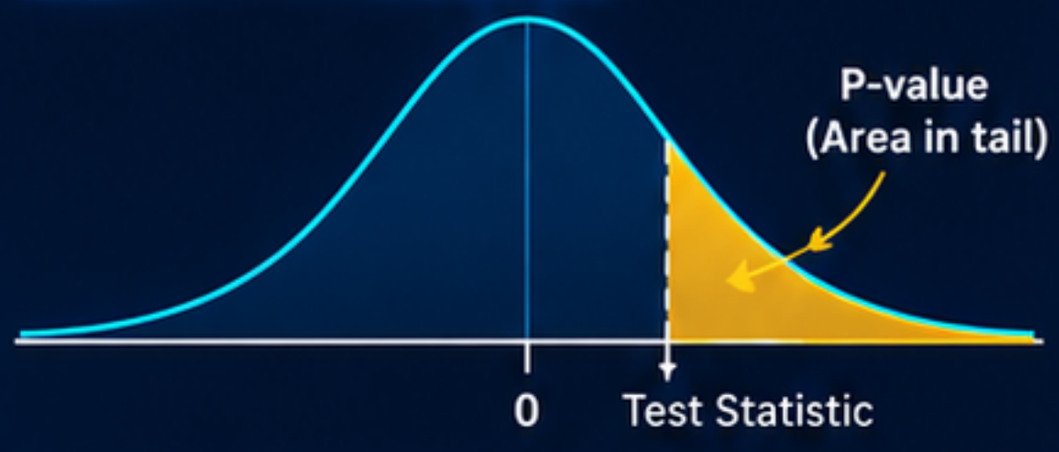
TOPICS WE WILL COVER

1. Introduction to Hypothesis Testing
2. Null Hypothesis (H_0)
3. Alternative Hypothesis (H_1)
4. One-Tailed vs Two-Tailed Test
5. Level of Significance (α)
6. Test Statistic
7. P-value
8. Critical Value Approach
9. Decision Rule
10. Type I Error (False Positive)
11. Type II Error (False Negative)
12. Power of a Test
13. Statistical Significance vs Practical Significance
14. Steps in Hypothesis Testing
15. Business Interpretation of Hypothesis Testing

HYPOTHESIS TESTING IN ACTION



P-VALUE CONCEPT



THE BIG PICTURE

State Hypotheses (H_0 vs H_1) → Collect and Analyze Data → Apply Statistical Test → Make a Decision (Reject or Fail to Reject H_0) → Draw Business Insights

TECH WITH SHIVANSH

Statistics for Data Analytics & Data Science

"Better Analysts Make a Smarter Tomorrow."

1 Introduction to Hypothesis Testing

Making Data-Driven Decisions, Not Just Guesses!

What is Hypothesis Testing?

A statistical method used to make decisions about a population based on sample data. It helps us check whether a claim or assumption about a population parameter is true or not.

> Hypothesis testing gives us a structured way to decide using evidence from data.

Key Points

- Uses sample data to test a claim about the population.
- Helps in decision making under uncertainty.
- Based on probability and statistical evidence.
- Widely used in real-world business and analytical problems.
- Does not prove a claim, but gives evidence to support or reject it.

Why Do We Need It?

- To validate business claims
- To compare performance before and after a change
- To test if a new strategy works
- To make data-driven decisions instead of assumptions
- To reduce risk in business decisions

Real-World Question
Is the new marketing campaign really increasing the conversion rate, or is the increase just by chance?

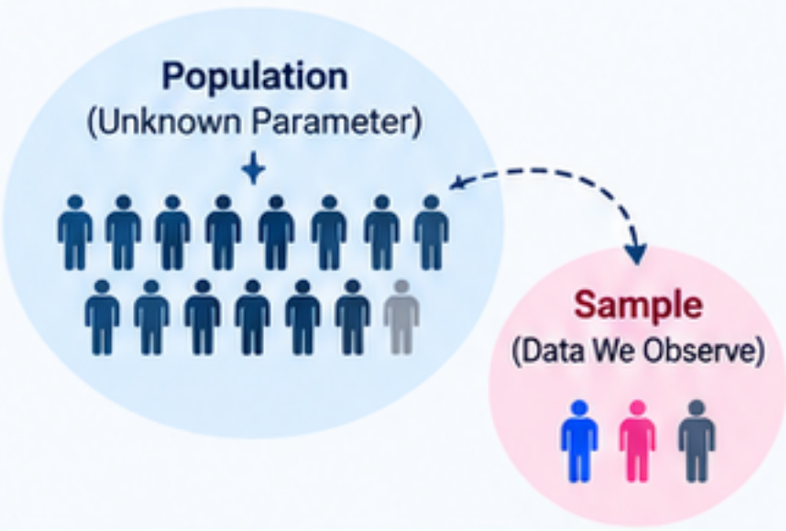
Real-Life Examples

E-commerce	Is the new website design improving conversion rate?
Finance	Is the average transaction value different after a new policy?
Healthcare	Is the new drug more effective than the old one?
Manufacturing	Is the average weight of products within acceptable limits?
HR Analytics	Is the average employee satisfaction different after a new work policy?

Basic Idea

Start with a Claim (Assumption)
↓
Collect Sample Data
↓
Analyze using Statistical Test
↓
Make a Decision (Reject or Fail to Reject)

Visual Representation



We use sample data to make an inference about the population.

Important Terminology

Term	Meaning
Population	Entire group of interest.
Sample	Subset of population.
Parameter	Numerical value describing a population (e.g., μ, p).
Statistic	Numerical value calculated from sample data (e.g., $\bar{x}$, $\bar{p}$).
Hypothesis	A claim or assumption about a population parameter.
Hypothesis Testing	A method to decide whether to reject or fail to reject a claim using sample data.

Example Scenario

A company claims that the average delivery time is 30 minutes.

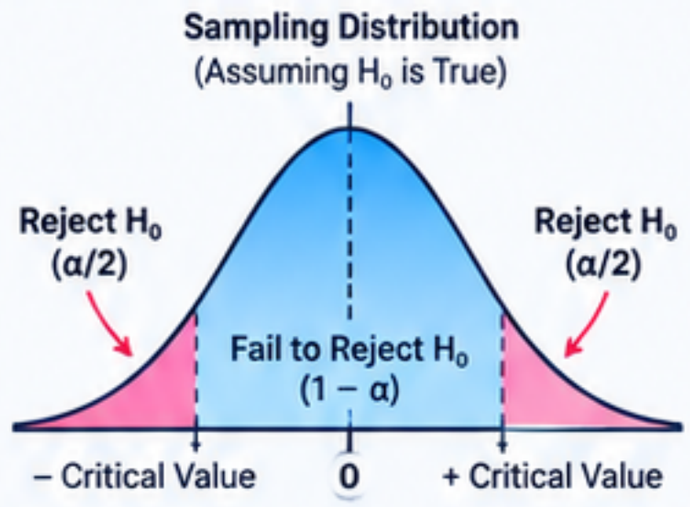
Is this claim true?

Set up a test:

- H_0: μ = 30 minutes (The claim is true)
- H_1: μ ≠ 30 minutes (The claim is not true)

We will use sample data to check if the evidence supports or contradicts this claim.

Graphical Intuition



If the sample result falls in the extreme region (tails), we reject the null hypothesis.

Key Takeaways

- Hypothesis testing helps us make decisions using data.
- We test a claim about a population parameter.
- It is widely used in business, healthcare, finance, and more.
- It doesn't prove a claim, but gives statistical evidence.
- Understanding the basics is crucial for advanced topics.

Next Topic

2. Null Hypothesis (H_0)
3. Alternative Hypothesis (H_1)
4. One-Tailed vs Two-Tailed Test
5. Level of Significance (α)
6. And more...

Quick Thought

"In data analytics, we don't just look at what happened, we ask — is it really happening or just by chance?"

2 Null Hypothesis (H_0)

Starting Point of Hypothesis Testing

What is Null Hypothesis (H_0)?

The null hypothesis (H_0) is a statement that there is no significant difference, no effect, or no change. It represents the status quo or a claim of no impact.

> H_0 assumes that any observed difference in the sample is due to random chance only.

Key Points

- Always contains equality (=, ≤, ≥).
- Represents the current belief or business claim.
- Assumes no effect or no difference.
- We try to reject H_0 using sample data and statistical evidence.
- If we fail to reject H_0, we do not prove it true, we just don't have enough evidence against it.

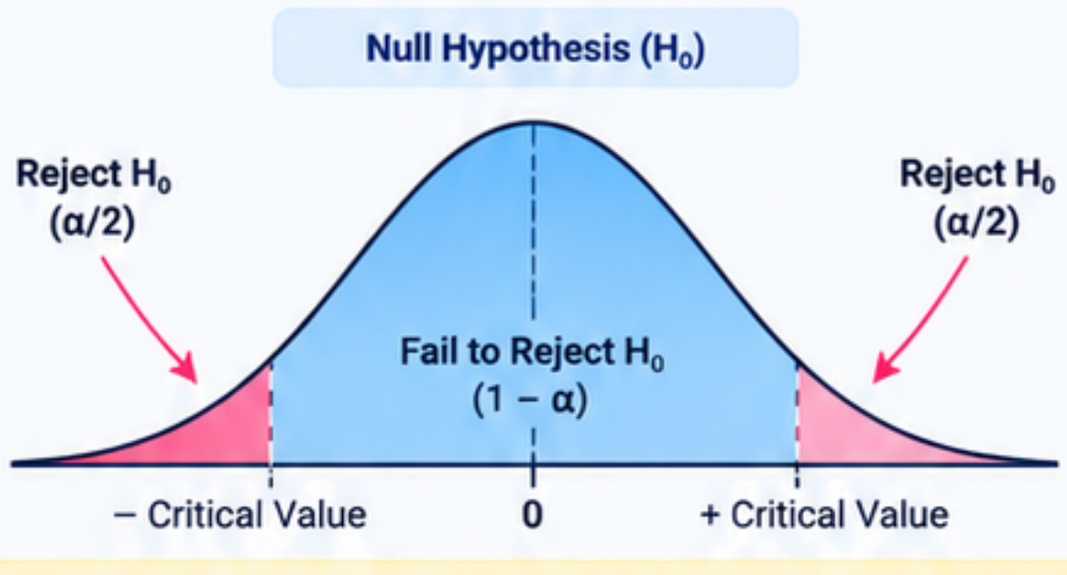
Examples of Null Hypothesis

1. H_0: $\mu = 500$ (The average weight is 500g)
2. H_0: $p = 0.20$ (The conversion rate is 20%)
3. H_0: $\mu_1 = \mu_2$ (No difference in means)
4. H_0: $\mu \leq 30$ (Average delivery time is not more than 30 minutes)
5. H_0: $\mu \geq 50$ (Average sales is at least 50)
6. H_0: $\mu_1 - \mu_2 = 0$ (No difference between two groups)
7. H_0: $p_1 = p_2$ (No difference in proportions)

Real-World Business Examples

Business Area	Null Hypothesis (H_0)	Meaning
E-commerce	H_0: $p = 0.05$	Conversion rate is 5% (no change)
Finance	H_0: $\mu = 1000$	Average transaction value is ₹1,000
Healthcare	H_0: $\mu_1 = \mu_2$	No difference in recovery time
Manufacturing	H_0: $\mu = 500$	Average product weight is 500g
HR Analytics	H_0: $\mu_1 = \mu_2$	No difference in average salary between two departments

Visual Representation



H_0 assumes the population parameter is equal to a specific value. We reject H_0 only if the sample result falls in the critical region (tails).

Types of Null Hypothesis

Equality Form (=)	Inequality Form (≤)	Inequality Form (≥)
H_0: $\mu = 50$	H_0: $\mu \leq 30$	H_0: $\mu \geq 70$
Used in most tests (two-tailed).	Used in right-tailed test.	Used in left-tailed test.

One-tailed vs Two-tailed H_0

Two-tailed Test

H_0: $\mu = 50$

Checks for difference in both directions (> or <).

One-tailed Test

H_0: $\mu \leq 30$

Checks for difference in one direction only (> or <).

Common Misconceptions

- H_0 is always true.
- Failing to reject H_0 means H_0 is true.
- H_0 contains "≠" sign.
- H_0 proves a claim.
- H_0 means there is no data variation.
- We can accept H_0.
- H_0 means no difference for individuals.

Step-by-Step Example

A company claims that the average delivery time is 30 minutes. We want to test this claim.

1. Define the Null Hypothesis

H_0: $\mu = 30$ minutes
(The claim is true)

2. Define the Alternative Hypothesis

H_1: $\mu \neq 30$ minutes
(The claim is not true)

Sample Data

Parameter	Value
Sample Mean ($\bar{x}$)	32
Sample Standard Deviation (s)	6
Sample Size (n)	36
Significance Level (α)	0.05

Hypotheses

H_0: $\mu = 30$
H_1: $\mu \neq 30$

Excel Calculation (Test Statistic)

1. Calculate Test Statistic (Z or t)

$$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Excel Formula
=(32-30)/(6/SQRT(36))
Result: 2.00

2. Find Critical Value (Two-tail, α = 0.05)

Z critical = ±1.96

Excel Formula
=NORM.S.INV(0.975)
Result: 1.96

3. Decision

Since |2.00| > 1.96 → Reject H_0

3 Alternative Hypothesis (H_1)

What We Want to Prove

What is Alternative Hypothesis (H_1)?

The alternative hypothesis (H_1) is a statement that contradicts the null hypothesis (H_0). It represents the claim that there is a significant effect, difference or relationship in the population.

> "H_1 is what we want to find evidence for in the data."

Key Points

- H_1 is always opposite to H_0.
- It represents the researcher's or business claim.
- It can be two-tailed or one-tailed.
- We reject H_0 in favour of H_1 when evidence is strong enough (p-value ≤ α).
- H_1 indicates the direction or presence of an effect.

Why is H_1 Important?

- Helps us test a real business claim.
- Shows whether a change or effect exists.
- Used to compare groups, test improvements and measure impact.
- Provides direction to the test (two-tailed or one-tailed).
- Essential for data-driven decisions.

Types of Alternative Hypothesis (H_1)

Two-tailed (≠)

Tests for any difference (increase or decrease).

$H_1: \mu \neq \mu_0$

Used when we don't know the direction.

Right-tailed (>)

Tests for an increase or greater value.

$H_1: \mu > \mu_0$

Used when we expect the value to be higher.

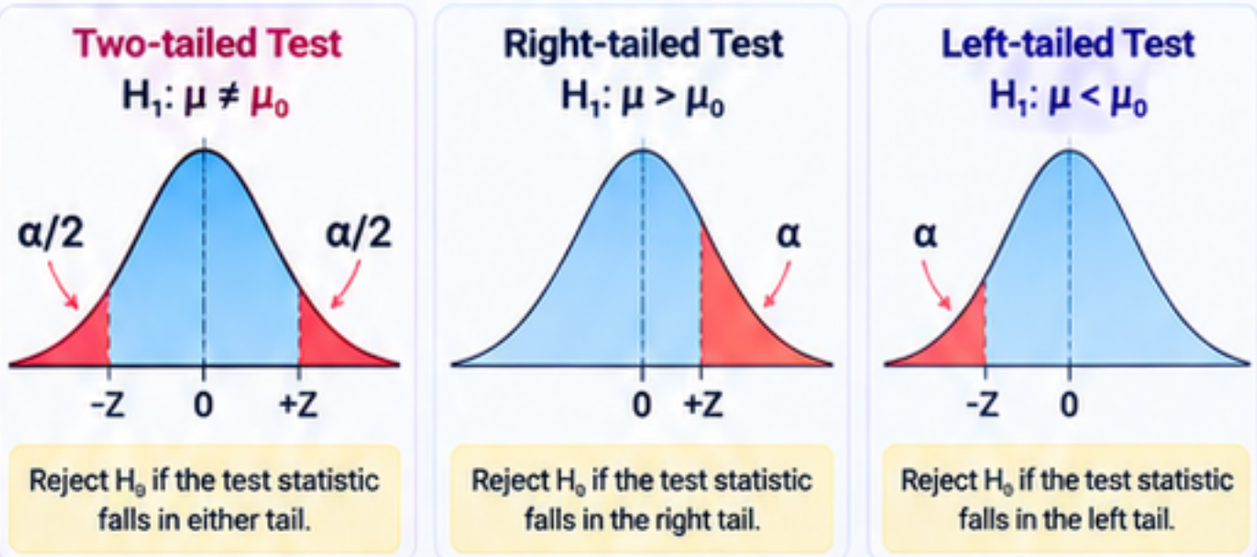
Left-tailed (<)

Tests for a decrease or smaller value.

$H_1: \mu < \mu_0$

Used when we expect the value to be lower.

Visual Representation



Real-World Business Examplees

Business Area	Business Question	Alternative Hypothesis (H_1)
E-commerce	Has the new website increased conversion rate?	$H_1: p > 0.05$
Marketing	Did the new campaign change the conversion rate?	$H_1: p \neq 0.05$
Finance	Is the average transaction value higher after new policy?	$H_1: \mu > 1000$
Manufacturing	Is the average defect rate lower after process improvement?	$H_1: p < 0.10$
Healthcare	Is the new drug more effective than the old one?	$H_1: \mu_1 > \mu_2$

Example Scenarios

1. **Two-tailed:** A company wants to check if the average delivery time is different from 30 minutes.
   $H_1: \mu \neq 30$
2. **Right-tailed:** A marketing team wants to check if the conversion rate has increased from 5%.
   $H_1: p > 0.05$
3. **Left-tailed:** A factory wants to check if the average product weight has decreased from 500g.
   $H_1: \mu < 500$
4. **Two-tailed:** HR wants to check if the average salary differs between two departments.
   $H_1: \mu_1 \neq \mu_2$

Step-by-Step Example (Right-tailed Test)

A company claims that the new campaign has increased the conversion rate from 5%. We test this claim.

1 **Define H_0:**	$H_0: p = 0.05$ (The conversion rate is 5%)
2 **Define H_1:**	$H_1: p > 0.05$ (The conversion rate is higher than 5%)
3 **Choose significance level:**	$\alpha = 0.05$
4 **Collect sample data:**	$n = 200$, Conversions = 15 → $\hat{p} = 0.075$
5 **Perform test and find p-value:**	p-value = 0.032
6 **Decision:**	Since p-value (0.032) < 0.05 → Reject H_0
7 **Conclusion:**	There is significant evidence that the conversion rate has increased.

Excel Calculation (Proportion Test)

Step	Excel Formula	Result
1	Sample proportion ($\hat{p}$)	=15/200	0.075
2	Standard error (SE)	=SQRT(0.05*(1-0.05)/200)	0.0154
3	Z statistic	=(0.075-0.05)/0.0154	1.62
4	P-value (right-tail)	=1-NORM.S.DIST(1.62,TRUE)	0.0526
5	Decision	Compare p-value with 0.05	Do not Reject H_0

Interpretation:

The p-value is 0.0526, which is slightly greater than 0.05, so we do not have enough evidence to conclude that the conversion rate has increased at 5% significance level.

Key Takeaways

- H_1 represents the claim we want to test.
- It can be two-tailed, right-tailed or left-tailed.
- It always contradicts H_0.
- The type of H_1 depends on the research question.
- We reject H_0 when p-value ≤ α.
- H_1 helps us make meaningful business decisions.

4 One-Tailed vs Two-Tailed Test

Choosing the Right Test for Your Research Question

What are One-Tailed and Two-Tailed Tests?

One-tailed and two-tailed tests are types of hypothesis tests that differ in the direction of the alternative hypothesis (H_1) and the rejection region.

> Choose the type of test based on your research question or business objective.

Key Differences

Aspect	Two-Tailed Test	One-Tailed Test
Alternative Hypothesis	Tests for any difference (≠)	Tests for an increase (>) or decrease (<)
Rejection Region	Both tails (α/2 in each tail)	One tail (all α in one tail)
Use Case	When direction is not specified	When direction is specified
Critical Value (at α = 0.05)	±1.96	+1.645 or −1.645

When to Use Which Test?

Use Two-Tailed Test (≠) when:

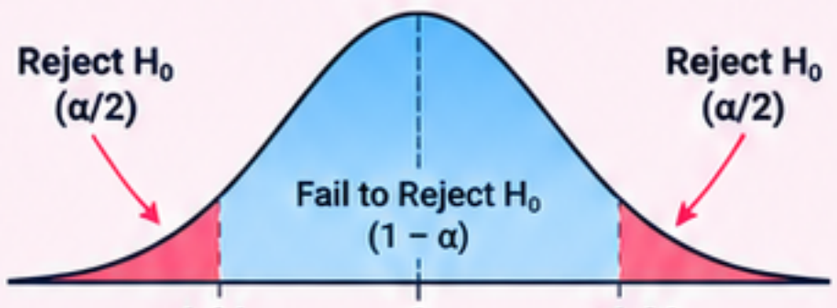
- You want to test if there is a difference (increase or decrease).
- No specific direction is mentioned.
- Common in academic research.
- **Example:** Is the average salary different from ₹50,000?

Use One-Tailed Test (> or <) when:

- You want to test only for an increase or a decrease.
- You have a specific direction based on business logic or prior research.
- Common in business and industry.
- **Example:** Has the conversion rate increased?

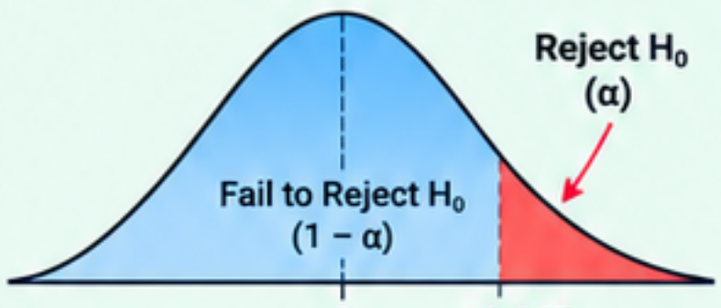
Visual Representation

Two-Tailed Test
$H_1: \mu \neq \mu_0$



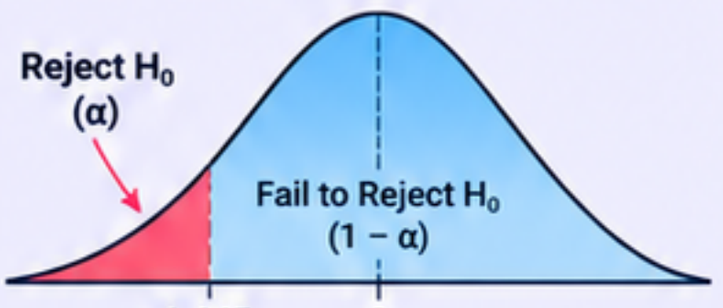
With α = 0.05, critical values are ±1.96 (α/2 = 0.025 in each tail).

Right-Tailed Test
$H_1: \mu > \mu_0$



With α = 0.05, critical value is +1.645 (all α in right tail).

Left-Tailed Test
$H_1: \mu < \mu_0$



With α = 0.05, critical value is −1.645 (all α in left tail).

Real-World Business Examples

Business Area	Research Question	Type of Test	Hypotheses
E-commerce	Has the new website changed the conversion rate?	Two-Tailed	$H_0: p = 0.05$ $H_1: p \neq 0.05$
Marketing	Has the new campaign increased the conversion rate?	Right-Tailed	$H_0: p = 0.05$ $H_1: p > 0.05$
Manufacturing	Has the average defect rate decreased after process improvement?	Left-Tailed	$H_0: p = 0.10$ $H_1: p < 0.10$
HR Analytics	Is the average employee satisfaction different from the previous year?	Two-Tailed	$H_0: \mu = \mu_0$ $H_1: \mu \neq \mu_0$

Numerical Examples with Graphs

Example 1: Two-Tailed Test (Mean)

A company wants to check if the average delivery time is different from 30 minutes.

- Sample mean ($\bar{X}$) = 32
- Sample standard deviation (s) = 6
- Sample size (n) = 36
- Significance level (α) = 0.05

Hypotheses:
$H_0: \mu = 30$
$H_1: \mu \neq 30$

Critical values: ±1.96 | Test statistic: 2.00 | Decision: Reject H_0

Example 2: Right-Tailed Test (Proportion)

A company wants to check if the conversion rate has increased from 5%.

- Sample proportion ($\bar{p}$) = 0.075
- Sample size (n) = 200
- Significance level (α) = 0.05

Hypotheses:
$H_0: p = 0.05$
$H_1: p > 0.05$

Critical value: +1.645 | Test statistic: 1.62 | Decision: Fail to Reject H_0

Excel Calculation (Critical Values)

Test Type	Excel Formula (α = 0.05)	Result
Two-Tailed	=NORM.S.INV(1-0.05/2)	±1.96
Right-Tailed	=NORM.S.INV(1-0.05)	1.645
Left-Tailed	=NORM.S.INV(0.05)	-1.645

Key Takeaways

- Two-tailed test checks for any difference.
- One-tailed test checks for a specific direction.
- Critical value depends on the test type and α.
- Choose the test based on the research question.
- Always align the test with business logic.

Quick Tip

If the question is "different," use two-tailed.
If the question has a specific direction (increase/decrease), use one-tailed.

5 Level of Significance (α)

How Much Evidence is Enough to Reject H_0?

What is Level of Significance (α)?

The level of significance (α) is the probability of rejecting the null hypothesis (H_0) when it is actually true.

> α represents the maximum risk we are willing to take of making a Type I error (False Positive).

Key Points

- Common values: 0.10, 0.05, 0.01
- Smaller α = stronger evidence required to reject H_0
- Larger α = easier to reject H_0 (but higher risk of Type I error)
- Usually chosen before collecting data
- Depends on business context and the consequence of wrong decision

Why is α Important?

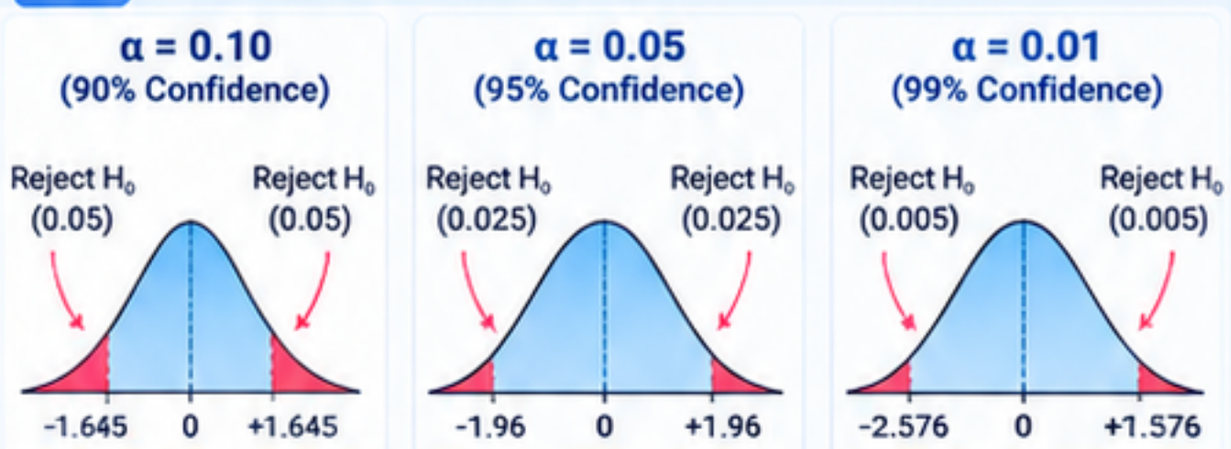
- Controls the risk of Type I error (False Positive)
- Determines the critical value
- Affects the rejection region
- Helps in decision making
- Should be chosen based on business impact (cost of wrong decision)

Common Significance Levels

α (Significance Level)	Confidence Level (1 − α)	Z Critical Value (Two-tailed)
0.10	90%	±1.645
0.05	95%	±1.960
0.01	99%	±2.576

Higher confidence level (smaller α) → Wider critical region and stricter rule to reject H_0.

Visual Representation



Real-World Business Examples

Business Area	Scenario	Why this α?
Healthcare	Testing a new drug effectiveness.	**Use α = 0.01** (because wrong decision can be life-threatening)
Manufacturing	Checking product defect rate.	**Use α = 0.05** (balance between risk and cost)
Marketing	Testing if new campaign increases conversion rate.	**Use α = 0.05** (common in business analysis)
Finance	Testing if fraud detection model is effective.	**Use α = 0.01** (high cost of wrong decision)
HR Analytics	Testing if average salary differs between two groups.	**Use α = 0.10** (exploratory analysis)

Effect of α on Critical Value and p-value

Significance Level (α)	Critical Value (Two-tailed)	Rejection Rule
0.10	±1.645	Reject H_0 if |test statistic| > 1.645
0.05	±1.960	Reject H_0 if |test statistic| > 1.960
0.01	±2.576	Reject H_0 if |test statistic| > 2.576

Smaller α → larger critical value → harder to reject H_0
Larger α → smaller critical value → easier to reject H_0

Relationship with p-value

- If p-value ≤ α → Reject H_0 (statistically significant result)
- If p-value > α → Fail to Reject H_0 (not statistically significant)

Excel Calculation of Critical Value

Test Type	Excel Formula (α = 0.05)	Result
Two-tailed (Z)	=NORM.S.INV(1-0.05/2)	1.96
Right-tailed (Z)	=NORM.S.INV(1-0.05)	1.645
Left-tailed (Z)	=NORM.S.INV(0.05)	-1.645

For t-distribution, use:
=T.INV.2T(α, degrees_of_freedom)

Example: Impact of α on Decision

A company tests if a new campaign increases the conversion rate.

- Test statistic (z) = 1.80
- p-value = 0.072

Significance Level (α)	Decision
0.10	Reject H_0 (1.80 > 1.645)
0.05	Fail to Reject H_0 (1.80 < 1.96)
0.01	Fail to Reject H_0 (1.80 < 2.576)

Key Takeaways

- α is the probability of Type I error (False Positive).
- Common values: 0.10, 0.05, 0.01.
- Smaller α requires strvdgenne.
- α should be chosen based on business context.
- Compare p-value with α to make a decision.
- α directly affects the critical value and rejection region.

6 Test Statistic

Converting Sample Evidence into a Number

What is Test Statistic?

A test statistic is a numerical value calculated from sample data that measures how far the sample result is from the null hypothesis value.

> "The test statistic tells us how extreme our sample result is if H_0 is true."

Key Points

- Calculated using sample data.
- Measures the difference between sample result and H_0 value.
- Larger absolute value → stronger evidence against H_0.
- Used to find p-value or compare with critical value.
- Different formula for mean, proportion and other tests (Z or t).
- Just a number, not a decision by itself.

Why is it Important?

- Helps us quantify the evidence against H_0.
- Used to calculate p-value.
- Shows how unusual the sample result is under H_0.
- Essential for decision making in hypothesis testing.
- Works for both Z-test and t-test.
- Commonly used in business and analytical problems.

Test Statistic Formulas

Z-test for Mean (σ Known)

$$z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

Where:
- $\bar{X}$ = sample mean
- μ_0 = hypothesized mean
- σ = population standard deviation

t-test for Mean (σ Unknown)

$$t = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}$$

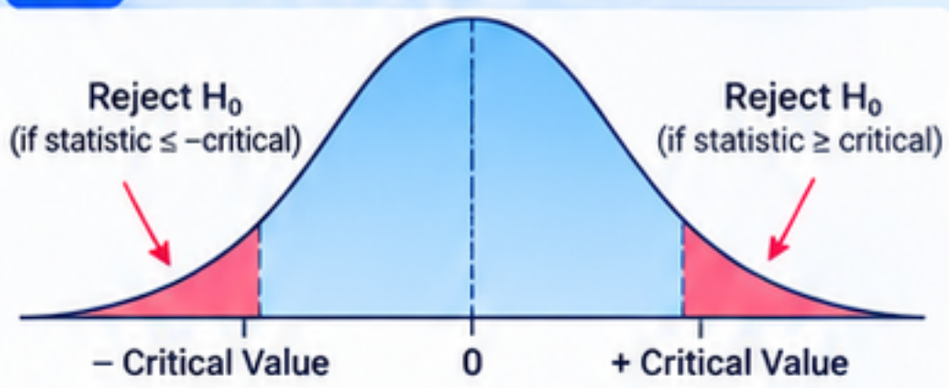
Where:
- $\bar{X}$ = sample mean
- μ_0 = hypothesized mean
- s = sample standard deviation
- n = sample size
- $df = n - 1$

Z-test for Proportion

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Where:
- $\hat{p}$ = sample proportion
- p_0 = hypothesized proportion
- n = sample size

Visual Representation



The test statistic shows where our sample result falls on the distribution under H_0. If it falls in the tail (rejection region), we reject H_0.

Example 1: Z-test for Mean (σ Known)

A company claims that the average delivery time is 30 minutes. A sample of 100 orders has a mean of 32 minutes with a population standard deviation of 6 minutes. Test at $\alpha = 0.05$ (two-tailed).

Parameter	Value
Sample Mean ($\bar{X}$)	32
Hypothesized Mean (μ_0)	30
Population SD (σ)	6
Sample Size (n)	100
Significance Level (α)	0.05

Calculation

$$z = \frac{32 - 30}{6/\sqrt{100}}$$

$$z = \frac{2}{0.6}$$

$$z = 3.33$$

Interpretation

Since $|3.33| > 1.96$, we reject H_0. There is sufficient evidence that the average delivery time is different from 30 minutes.

Example 2: t-test for Mean (σ Unknown)

A company wants to test if the average salary is more than ₹50,000. A sample of 25 employees has a mean salary of ₹52,000 and a standard deviation of ₹6,000. Test at $\alpha = 0.05$ (right-tailed).

Parameter	Value
Sample Mean ($\bar{X}$)	52000
Hypothesized Mean (μ_0)	50000
Sample SD (s)	6000
Sample Size (n)	25
Degrees of Freedom (df)	24
Significance Level (α)	0.05

Calculation

$$t = \frac{52000 - 50000}{6000/\sqrt{25}}$$

$$t = \frac{2000}{1200}$$

$$z = 1.67$$

Interpretation

Critical value (right-tail, df = 24) ≈ 1.711. Since $1.67 < 1.711$, we fail to reject H_0. There is not enough evidence that the average salary is more than ₹50,000.

Example 3: Z-test for Proportion

A marketing team claims that the conversion rate is 20%. In a sample of 500 visitors, 120 made a purchase. Test at $\alpha = 0.05$ (two-tailed).

Parameter	Value
Sample Proportion ($\hat{p}$)	0.24
Hypothesized Proportion (p_0)	0.20
Sample Size (n)	500
Significance Level (α)	0.05

Calculation

$$z = \frac{0.24 - 0.20}{\sqrt{\frac{0.20(1-0.20)}{500}}}$$

$$z = \frac{0.04}{0.01789}$$

$$z = 2.24$$

Interpretation

- Since $|2.24| > 1.96$, **we reject H_0.**
- There is sufficient **evidence that the** conversion rate is different from 20%.

Excel Functions for Test Statistic

Test Type	Excel Formula	Example
Z-test for Mean (σ known)	=(B2-B3)/(B4/SQRT(B5))	=(32-30)/(6/SQRT(100)) = 3.33
t-test for Mean (σ unknown)	=(B2-B3)/(B4/SQRT(B5))	=(52000-50000)/(6000/SQRT(25)) = 1.67
Z-test for Proportion	=(B2-B3)/SQRT(B3*(1-B3)/B4)	=(0.24-0.20)/SQRT(0.20*(1-0.20)/500) = 2.24

Tip: Use absolute value (ABS) and compare with critical value or calculate p-value using NORM.S.DIST() or T.DIST().

7 P-value

Measuring the Strength of Evidence

What is a P-value?

The p-value is the probability of observing a test statistic at least as extreme as the one calculated from the sample data, assuming that the null hypothesis (H_0) is true.

> "P-value tells us how likely our sample result is, if H_0 is true."

Key Points

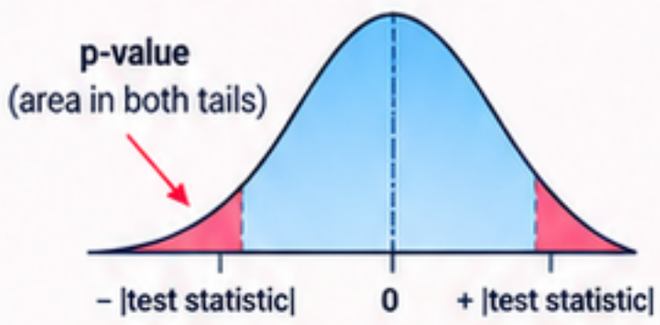
- P-value is always between 0 and 1.
- Smaller p-value → stronger evidence against H_0.
- If p-value ≤ α, we reject H_0.
- If p-value > α, we fail to reject H_0.
- P-value does not tell us the size or importance of the effect.
- A very small p-value means the result is unlikely to have occurred by random chance (if H_0 is true).

Why is P-value Important?

- Helps us make an objective decision.
- Quantifies the evidence against H_0.
- Avoids using only the critical value.
- Widely used in business, research and analytics.
- Works for both Z-test and t-test.
- Helps compare results across different tests.
- Smaller p-value = stronger evidence, but we also need to check practical significance.

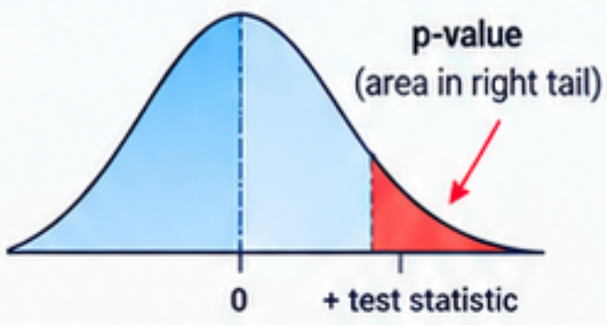
Visual Explanation of P-value

Two-tailed Test



P-value is the total area in both tails beyond the test statistic (more extreme values).

Right-tailed Test



P-value is the area in the right tail beyond the test statistic.

P-value vs Level of Significance

Aspect	P-value	Level of Significance (α)
Meaning	Probability of observing the result (or more extreme) if H_0 is true.	Pre-decided threshold for decision making.
Value Range	0 to 1	Common values: 0.10, 0.05, 0.01
Calculated or Fixed	Calculated from sample data.	Fixed before analysis.
Use	Compare with α to make a decision.	Standard for rejecting H_0.
Interpretation	Smaller p-value = stronger evidence against H_0.	Smaller α = stricter criterion to reject H_0.

Decision Rule using P-value

If p-value ≤ α

Reject H_0

There is sufficient evidence against the null hypothesis.

If p-value > α

Fail to Reject H_0

There is not enough evidence against the null hypothesis.

P-value and Evidence Strength

P-value Range	Strength of Evidence
$p \geq 0.10$	No significant evidence
$0.05 \leq p < 0.10$	Weak evidence
$0.01 \leq p < 0.05$	Moderate evidence
$0.001 \leq p < 0.01$	Strong evidence
$p < 0.001$	Very strong evidence

Common Misconceptions

- P-value is the probability that H_0 is true.
- A small p-value always means a large or important effect.
- P-value tells us the probability of getting the exact same result.
- If p-value > 0.05, it proves H_0 is true.
- P-value depends on sample size (mechanically smaller with larger n).
- Statistical significance always means practical significance.

Numerical Examples with Excel Calculation

Example 1: Two-tailed Z-test (Mean)

A company wants to check if the average delivery time is different from 30 minutes. Sample mean = 32, σ = 6, n = 100, α = 0.05

Step 1: Calculate test statistic (z)

$$z = \frac{32 - 30}{6/\sqrt{100}} = \frac{2}{0.6} = \mathbf{3.33}$$

Step 2: Calculate p-value (two-tailed)

Excel: `=2*(1-NORM.S.DIST(3.33,TRUE))`

Result: **0.0009**

Decision

p-value (0.0009) < 0.05
→ **Reject H_0**
The average delivery time is different from 30 minutes.

Example 2: Right-tailed t-test (Mean)

A marketing team wants to check if the average salary is more than ₹50,000. Sample mean = 52,000, s = 6,000, n = 25, α = 0.05

Step 1: Calculate test statistic (t)

$$t = \frac{52000 - 50000}{6000/\sqrt{25}} = \frac{2000}{1200} = \mathbf{1.67}$$

Step 2: Calculate p-value (right-tailed)

Excel: `=1-T.DIST(1.67,24,TRUE)`

Result: **0.0547**

Decision

p-value (0.0547) > 0.05
→ **Fail to Reject H_0**
Not enough evidence that average salary is more than ₹50,000.

Example 3: Left-tailed Z-test (Proportion)

A factory wants to check if the defect rate has decreased from 10%. Sample proportion = 0.07, n = 400, α = 0.05

Step 1: Calculate test statistic (z)

$$z = \frac{0.07 - 0.10}{\sqrt{0.10(0.90)/400}} = \frac{-0.03}{0.015} = \mathbf{-2.00}$$

Step 2: Calculate p-value (left-tailed)

Excel: `=NORM.S.DIST(-2.00,TRUE)`

Result: **0.0228**

Decision

p-value (0.0228) < 0.05
→ **Reject H_0**
There is sufficient evidence that the defect rate has decreased from 10%.

8 Critical Value Approach

Using Critical Values to Make a Decision

What is Critical Value Approach?

The critical value approach is a method of hypothesis testing where we compare the test statistic with a critical value obtained from the sampling distribution at a chosen significance level (α).

> "If our test statistic falls in the rejection region (beyond the critical value), we reject the null hypothesis (H_0)."

Key Points

- Critical value depends on α and the type of test (one-tailed or two-tailed).
- We compare the test statistic with the critical value.
- If test statistic > critical value (or < −critical value), reject H_0.
- Critical value approach gives the rejection region before looking at the data.
- Widely used in Z-test and t-test.

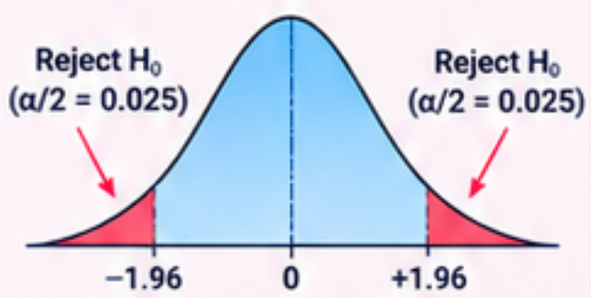
Why is it Important?

- Helps us clearly identify the rejection region.
- Provides a fixed threshold for decision making.
- Useful when we want to understand the cutoff point for significance.
- Widely used in business, research and analytics.
- Complements the p-value approach (both give the same decision).

Rejection Region and Critical Values

Two-tailed Test (α = 0.05)

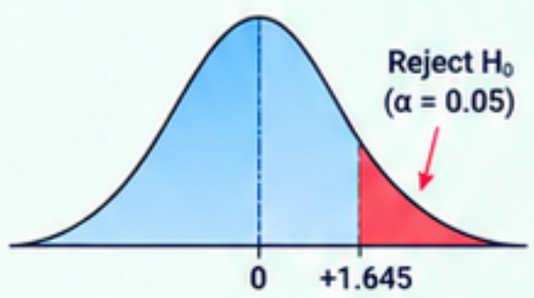
Reject H_0 if |test statistic| > critical value



Critical values: ±1.96
(α/2 = 0.025 in each tail)

Right-tailed Test (α = 0.05)

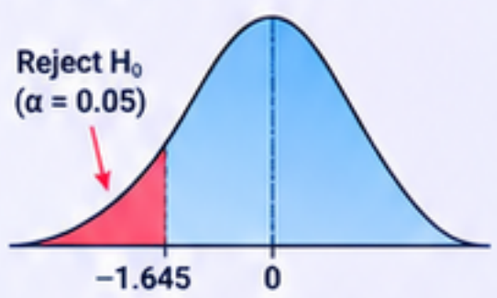
Reject H_0 if test statistic > critical value



Critical value: +1.645
(all α in right tail)

Left-tailed Test (α = 0.05)

Reject H_0 if test statistic < critical value



Critical value: −1.645
(all α in left tail)

Common Critical Values (Z)

Significance Level (α)	Two-tailed (±z)	Right-tailed (z)	Left-tailed (−z)
0.10	±1.645	1.282	−1.282
0.05	±1.960	1.645	−1.645
0.01	±2.576	2.326	−2.326

Smaller α → larger critical value → stricter requirement to reject H_0.

Steps in Critical Value Approach

1. State the hypotheses (H_0 and H_1).
2. Choose the significance level (α).
3. Identify the type of test (two-tailed, right-tailed or left-tailed).
4. Find the critical value from Z or t distribution.
5. Calculate the test statistic using sample data.
6. Compare the test statistic with the critical value.
7. Make the decision and interpret the result in a business context.

Decision Rule Summary

Type of Test	Rejection Rule (Critical Value Approach)
Two-tailed	Reject H_0 if |test statistic| > $z_{\alpha/2}$ (e.g., |z| > 1.96 for α = 0.05)
Right-tailed	Reject H_0 if test statistic > z_α (e.g., z > 1.645 for α = 0.05)
Left-tailed	Reject H_0 if test statistic < $-z_\alpha$ (e.g., z < −1.645 for α = 0.05)

Excel Formulas for Critical Values

Type of Test	Excel Formula (α = 0.05)
Two-tailed (Z)	=NORM.S.INV(1-0.05/2) (returns 1.96)
Right-tailed (Z)	=NORM.S.INV(1-0.05) (returns 1.645)
Left-tailed (Z)	=NORM.S.INV(0.05) (returns -1.645)
Two-tailed (t)	=T.INV.2T(0.05, df) (e.g., df = 24)
Right-tailed (t)	=T.INV(0.05, df)
Left-tailed (t)	=-T.INV(0.05, df)

Numerical Examples

Example 1: Two-tailed Z-test (Mean)

A company wants to check if the average delivery time is different from 30 minutes. Sample mean = 32, $\sigma = 6$, $n = 100$, $\alpha = 0.05$.

1. **Hypotheses:** $H_0: \mu = 30$, $H_1: \mu \neq 30$
2. **Test statistic:** $$z = \frac{32 - 30}{6/\sqrt{100}} = \frac{2}{0.6} = \mathbf{3.33}$$
3. **Critical value:** ±1.96
4. **Decision:** |3.33| > 1.96 → Reject H_0
5. **Conclusion:** Average delivery time is significantly different from 30 minutes.

Example 2: Right-tailed t-test (Mean)

A company wants to check if the average salary is more than ₹50,000. Sample mean = 52,000, $s = 6{,}000$, $n = 25$, $\alpha = 0.05$.

1. **Hypotheses:** $H_0: \mu = 50{,}000$, $H_1: \mu > 50{,}000$
2. **Test statistic:** $$t = \frac{52000 - 50000}{6000/\sqrt{25}} = \frac{2000}{1200} = \mathbf{1.67}$$
3. **Critical value (df = 24):** 1.711
4. **Decision:** 1.67 < 1.711 → Fail to Reject H_0
5. **Conclusion:** Not enough evidence that the average salary is more than ₹50,000.

Example 3: Left-tailed Z-test (Proportion)

A factory wants to check if the defect rate has decreased from 10%. Sample proportion = 0.07, $n = 400$, $\alpha = 0.05$.

1. **Hypotheses:** $H_0: p = 0.10$, $H_1: p < 0.10$
2. **Test statistic:** $$z = \frac{0.07 - 0.10}{\sqrt{0.10(0.90)/400}} = \frac{-0.03}{0.015} = \mathbf{-2.00}$$
3. **Critical value:** −1.645
4. **Decision:** −2.00 < −1.645 → Reject H_0
5. **Conclusion:** Defect rate has significantly decreased from 10%.

9 Decision Rule

How to Conclude in Hypothesis Testing

What is Decision Rule?

The decision rule tells us when to reject or fail to reject the null hypothesis (H_0) based on the test statistic, critical value, or p-value.

> "The decision rule converts our statistical calculation into a clear conclusion."

Key Points

- We can use either p-value approach or critical value approach.
- If p-value ≤ α → Reject H_0.
- If p-value > α → Fail to Reject H_0.
- If test statistic > critical value (or < −critical value) → Reject H_0.
- Decision rule depends on the type of test (one-tailed or two-tailed).
- Always connect the decision with business context.

Why is it Important?

- Helps us make a clear and objective decision.
- Avoids confusion after calculations.
- Ensures consistency in hypothesis testing.
- Widely used in business, research and data analytics.
- Helps translate statistical results into meaningful business insights.

Decision Rule Using p-value Approach

If p-value ≤ α

Reject H_0

- There is sufficient evidence against the null hypothesis.
- The result is statistically significant.
- We accept the alternative hypothesis (H_1).

If p-value > α

Fail to Reject H_0

- There is not enough evidence against the null hypothesis.
- The result is not statistically significant.
- We do not have strong evidence to support H_1.

Decision Rule Using Critical Value Approach

If test statistic falls in rejection region

Reject H_0

- For two-tailed test: |test statistic| > critical value
- For right-tailed test: test statistic > critical value
- For left-tailed test: test statistic < −critical value

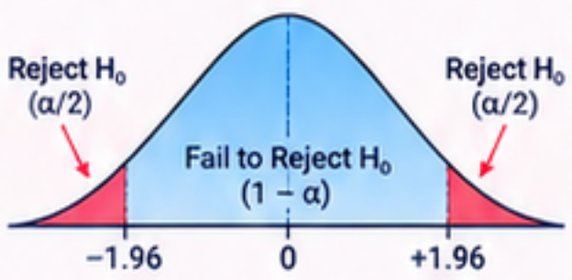
If test statistic does not fall in rejection region

Fail to Reject H_0

- For two-tailed test: |test statistic| ≤ critical value
- For right-tailed test: test statistic ≤ critical value
- For left-tailed test: test statistic ≥ −critical value

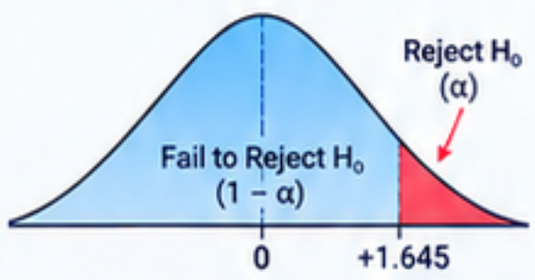
Visual Representation

Two-tailed Test (Reject H_0 in both tails)



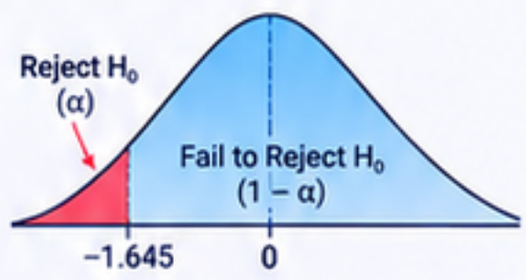
Reject H_0 if |test statistic| > 1.96 (at α = 0.05)

Right-tailed Test (Reject H_0 in right tail)



Reject H_0 if test statistic > 1.645 (at α = 0.05)

Left-tailed Test (Reject H_0 in left tail)



Reject H_0 if test statistic < −1.645 (at α = 0.05)

Common Significance Levels and Critical Values (Z)

α	Two-tailed	Right-tailed	Left-tailed
0.10	±1.645	1.282	−1.282
0.05	±1.960	1.645	−1.645
0.01	±2.576	2.326	−2.326

Smaller α → Larger critical value → Harder to reject H_0 (more strict).

Numerical Examples

Example 1: Two-tailed Z-test (Mean)

A company tests if the average delivery time is different from 30 minutes. Sample mean = 32, σ = 6, n = 100, α = 0.05.

1. **Test statistic:** $z = \frac{32 - 30}{6/\sqrt{100}} = \frac{2}{0.6} = 3.33$
2. **Critical value:** ±1.96
3. **Decision:** |3.33| > 1.96 → Reject H_0
4. **Conclusion:** Average delivery time is significantly different from 30 minutes.

Example 2: Right-tailed t-test (Mean)

A company tests if the average salary is more than ₹50,000. Sample mean = 52,000, s = 6,000, n = 25, α = 0.05.

1. **Test statistic:** $t = \frac{52000 - 50000}{6000/\sqrt{25}} = \frac{2000}{1200} = 1.67$
2. **Critical value (df = 24):** 1.711
3. **Decision:** 1.67 < 1.711 → Fail to Reject H_0
4. **Conclusion:** Not enough evidence that the average salary is more than ₹50,000.

Example 3: Left-tailed Z-test (Proportion)

A factory tests if the defect rate has decreased from 10%. Sample proportion = 0.07, n = 400, α = 0.05.

1. **Test statistic:** $z = \frac{0.07 - 0.10}{\sqrt{0.10(0.90)/400}} = \frac{-0.03}{0.015} = -2.00$
2. **Critical value:** −1.645
3. **Decision:** −2.00 < −1.645 → Reject H_0
4. **Conclusion:** Defect rate has significantly decreased from 10%.

Key Takeaways

- Decision rule can be based on p-value or critical value.
- Reject H_0 if p-value ≤ α or test statistic falls in the rejection region.
- Fail to reject H_0 if p-value > α or test statistic does not fall in the rejection region.
- Always interpret the decision in a real business context.

Quick Tips

- Always mention the significance level (α).
- Clearly state the decision (Reject or Fail to Reject H_0).
- Mention the reason (p-value or test statistic vs critical value).
- Give a business interpretation of the result.

10 Type I Error (False Positive)

Rejecting a True Null Hypothesis

What is Type I Error?

Type I error occurs when we reject the null hypothesis (H_0) even though it is actually true.

> "It means we conclude that there is an effect or difference when in reality there is none."

Key Points

- Type I error = False Positive
- Probability of Type I error is α (level of significance).
- If α = 0.05, there is a 5% chance of committing a Type I error.
- Smaller α reduces the risk of Type I error (but increases Type II error risk).
- We control the risk of Type I error by choosing an appropriate significance level.

Why is it Important?

- Helps us understand the risk of wrongly detecting an effect.
- Important in business decisions where false alarms can be costly.
- Choosing α depends on the consequences of a false positive.
- Common in medical, financial, marketing, and manufacturing studies.
- Allows better decision-making and risk management.

Visual Representation

When H_0 is actually true, the area in the rejection region (α) represents the probability of committing a Type I error.

Type I Error in a Contingency Table

Actual Situation	Decision	
	Reject H_0	Fail to Reject H_0
H_0 is True (No real effect)	**Type I Error** (False Positive) Probability = α	Correct Decision (True Negative) Probability = 1 – α
H_0 is False (Real effect exists)	Correct Decision (True Positive) Probability = 1 – β	**Type II Error** (False Negative) Probability = β

Real-World Business Examples

Business Area	Example	Type I Error (False Positive)
Healthcare	Concluding a new drug is effective when it is not.	Approving an ineffective drug (patient risk + cost).
Manufacturing	Detecting a defect in products when none actually exists.	Stopping production unnecessarily (time and cost loss).
Marketing	Concluding a new ad campaign increases sales when it does not.	Investing in a campaign that gives no real benefit.
Finance	Detecting unusual fraud activity when it is normal.	Unnecessary investigation and compliance cost.
HR Analytics	Concluding a training program improves performance when it does not.	Continuing a program that has no real impact.

Effect of α on Type I Error

Significance Level (α)	Probability of Type I Error	Interpretation
0.10	10%	Higher risk of false positive
0.05	5%	Commonly used in practice
0.01	1%	Lower risk of false positive

Smaller α → lower chance of Type I error but higher chance of Type II error (β).

Choosing α in Different Contexts

Context	Recommended α	Reason
Healthcare (life-critical)	0.01	Minimize false positive (patient safety).
Manufacturing (quality check)	0.05	Balance risk and cost.
Marketing (A/B testing)	0.05 or 0.10	False positive less serious.
Finance (fraud detection)	0.01 or 0.05	High cost of false alarm.
HR Analytics	0.05	Standard practice.

Numerical Example with Excel Calculation

A company claims that a new training program increases employee productivity. We test at α = 0.05 (two-tailed). We assume H_0 is actually true (no increase).

Parameter	Value
Sample Mean ($\bar{X}$)	50
Hypothesized Mean (μ_0)	50
Population SD (σ)	8
Sample Size (n)	64
Significance Level (α)	0.05

Calculation

Test statistic (Z):

$$z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{50-50}{8/\sqrt{64}} = \frac{0}{1} = 0$$

$$z = 0.00$$

Critical values and decision

For α = 0.05 (two-tailed):
Critical values = ±1.96
Since |0.00| < 1.96, we fail to reject H_0.

Interpretation

In this case, we do not make a Type I error because we did not reject H_0 (which is actually true). If the test statistic had fallen in the rejection region (|z| > 1.96), we would have committed a Type I error with probability α = 0.05 (5%).

Key Takeaways

- Type I error = rejecting a true H_0 (False Positive).
- Probability of Type I error = α.
- Smaller α reduces Type I error but increases Type II error.
- Choice of α depends on business context and risk tolerance.
- Always consider the real-world consequences of a false positive.
- Clearly communicate the risk of Type I error in decision-making.

11 Type II Error (False Negative)

Failing to Reject a False Null Hypothesis

What is Type II Error?

Type II error occurs when we fail to reject the null hypothesis (H_0) when it is actually false.

> "It means we conclude that there is no effect or difference when in reality an effect actually exists."

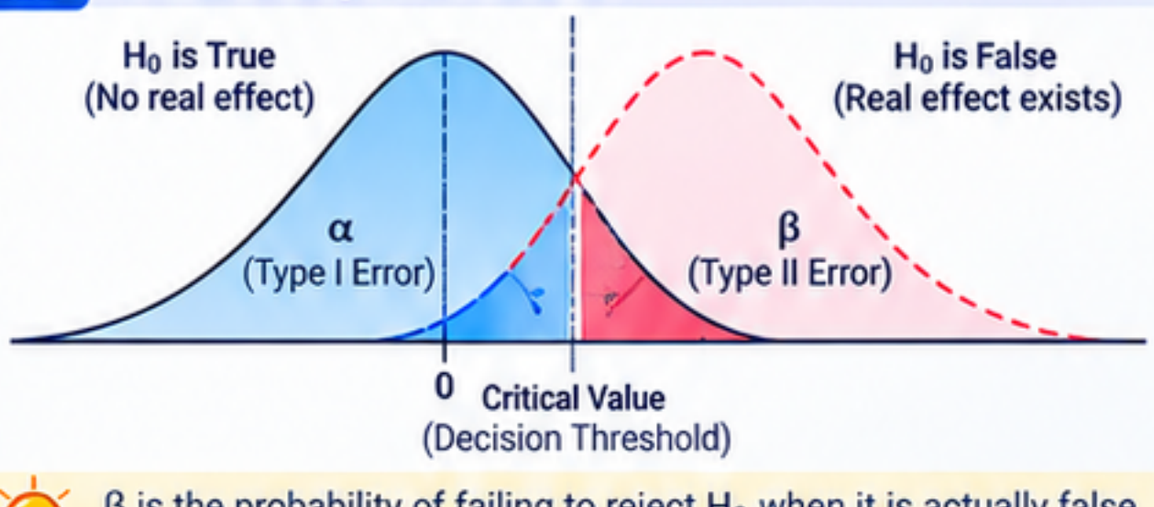
Key Points

- Type II error = False Negative
- Probability of Type II error is β (beta).
- If β = 0.20, there is a 20% chance of missing a real effect.
- Smaller β reduces the risk of Type II error.
- Increasing sample size (n) reduces β.
- Higher significance level (α) increases β.
- Related to the power of a test (Power = 1 − β).

Why is it Important?

- Helps us understand the risk of missing a real effect.
- Can lead to wrong business decisions (e.g., not launching a profitable campaign).
- Important in healthcare, safety, finance, and quality control.
- A high β means low power of the test.
- We must balance Type I and Type II errors based on business consequences.

Visual Representation



β is the probability of failing to reject H_0 when it is actually false (i.e., missing a real effect).

Type I Error vs Type II Error

Aspect	Type I Error (α)	Type II Error (β)
Definition	Rejecting a true H_0 (False Positive)	Failing to reject a false H_0 (False Negative)
Probability	α (level of significance)	β
Decision	Conclude there is an effect when there is none	Conclude there is no effect when an effect actually exists
Consequence	False alarm (unnecessary action)	Miss a real opportunity (lose benefit)
How to Reduce	Choose smaller α	Increase sample size (n), increase α, or use a more powerful test
Related to	Significance level (α)	Power (1 − β)

Real-World Business Examples

Business Area	Example	Type II Error (False Negative)
Healthcare	A new drug is effective, but the test fails to detect its effectiveness.	Continuing with an ineffective treatment and missing a potential life-saving drug.
Marketing	A new marketing campaign actually increases sales, but the test shows no difference.	Missing a profitable campaign opportunity.
Manufacturing	A defect exists in production, but the test fails to detect it.	Shipping defective products to customers.
Finance	A fraud detection system is effective, but the test fails to identify it.	Continuing fraud activity, leading to financial losses.
HR Analytics	A training program improves performance, but the test shows no effect.	Missing an opportunity to invest in a beneficial program.

Power of a Test

Power is the **probability of correctly** rejecting a false null hypothesis (detecting a real effect).

$$\text{Power} = 1 - \beta$$

- Higher power → lower chance of Type II error.
- Commonly used power values: 0.80 (80%) or 0.90 (90%).
- Power increases with larger sample size (n).
- Power also increases with larger effect size.
- There is a trade-off between α and β (lower α increases β).

Relationship between α, β and Power

Parameter	Meaning	Common Values
α	Probability of Type I error (False Positive)	0.10, 0.05, 0.01
β	Probability of Type II error (False Negative)	Varies (e.g., 0.20)
Power (1 − β)	Probability of detecting a real effect	0.80, 0.90

Numerical Example with Excel Calculation

A company claims that a new training program increases employee productivity. We test at α = 0.05 and want to understand β and power. Assume the true mean after training is 55, but our sample shows 52. σ = 8, n = 64.

Parameter	Value
Hypothesized Mean (μ_0)	50
True Mean (μ_1)	55
Sample Mean ($\bar{X}$)	52
Population SD (σ)	8
Sample Size (n)	64
Significance Level (α)	0.05

Calculation (Z-test)

Standard Error (SE) $= \frac{\sigma}{\sqrt{n}} = \frac{8}{\sqrt{64}} = 1$

Test statistic: $z = \frac{\bar{X} - \mu_0}{SE} = \frac{52 - 50}{1} = 2.00$

Critical value (two-tailed, α = 0.05): ±1.96

Decision: 2.00 > 1.96 → Reject H_0

In this example, we correctly detected the effect. But if the sample mean was smaller (e.g., 51), we might fail to reject H_0 even though the true mean is 55 → this would be a Type II error (β).

Factors Affecting Type II Error (β)

Factor	Effect on β	Explanation
Sample size (n)	↓ Decreases	Larger n gives more precise estimates.
Effect size (difference)	↓ Decreases	Larger real difference is easier to detect.
Significance level (α)	↑ Increases	Higher α increases β (trade-off with Type I error).
Variability (σ)	↑ Increases	More variability makes it harder to detect the effect.
Test power	↓ Decreases (when power ↑)	Higher power reduces β.

Key Takeaways

- Type II error occurs when we fail to reject a false H_0 (false Negative).
- Probability of Type II error is β.
- Power = 1 − β.
- Larger sample size and larger effect size reduce β.
- Higher variability and smaller α increase β.
- Type I and Type II errors must be balanced based on business impact.
- Use power analysis to plan sample size for important studies.

12 Power of a Test

Probability of Detecting a Real Effect

What is Power of a Test?

Power of a test is the probability of correctly rejecting the null hypothesis (H_0) when it is actually false (i.e., detecting a real effect).

> "Power tells us how good our test is at finding a real effect when it truly exists."

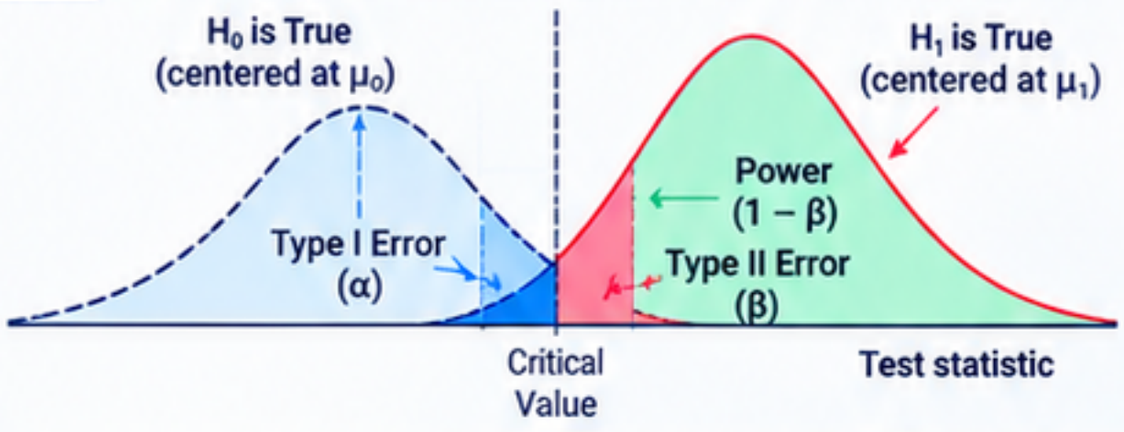
Key Points

- Power = $1 - \beta$
- Ranges between 0 and 1 (or 0% to 100%).
- Higher power → lower chance of Type II error (β).
- Common target power values: 0.80 (80%) or 0.90 (90%).
- Power depends on:
  - Effect size
  - Sample size (n)
  - Significance level (α)
  - Variability (σ)

Why is it Important?

- Helps us detect real business effects (differences, improvements).
- Reduces the risk of missing important opportunities.
- Important in A/B testing, clinical trials, marketing, quality control, etc.
- Guides us in choosing the right sample size.
- Low power can lead to false negatives (Type II error).
- Widely used in research, business analytics, and decision-making.

Visual Representation of Power

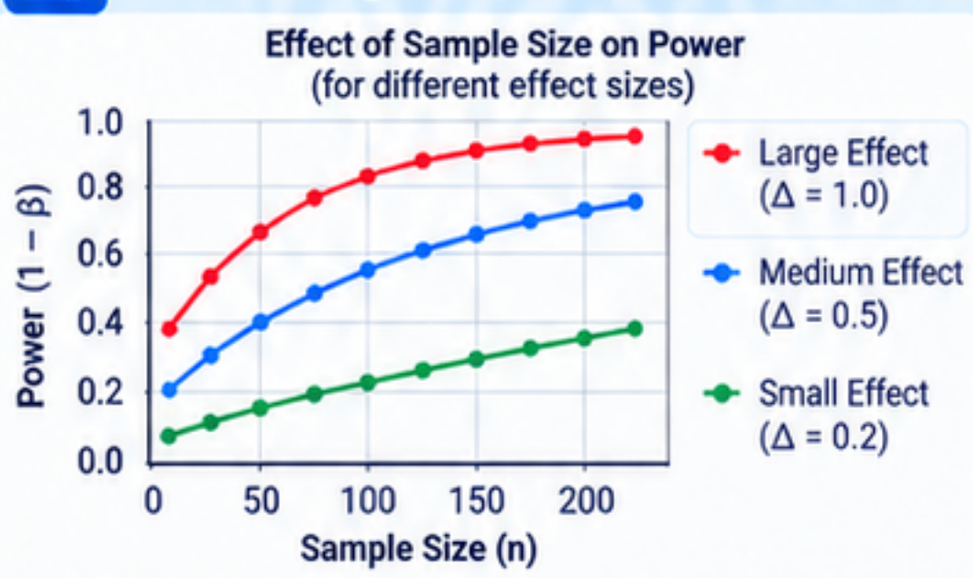


Power ($1 - \beta$) is the area under the alternative distribution (H_1) beyond the critical value. It represents the probability of correctly detecting a real effect.

Factors Affecting the Power of a Test

Factor	Effect on Power	Explanation
Effect Size (Δ)	Increases	Larger true difference between groups → easier to detect → higher power.
Sample Size (n)	Increases	Larger sample size → reduces standard error → higher power.
Significance Level (α)	Increases (when α is larger)	A higher α (e.g., 0.10) → larger rejection region → higher power (but also higher Type I error risk).
Variability (σ)	Decreases (when σ is larger)	Higher variability → larger standard error → lower power.

Power vs Significance Level and Effect Size



Common Power Values

Power (1 – β)	Type II Error (β)	Usage
0.50	0.50	Too low (not recommended)
0.70	0.30	Minimum acceptable (in some fields)
0.80	0.20	Commonly used (standard)
0.90	0.10	More stringent (important studies)
0.95	0.05	High power (critical applications)

A power of 80% means there is an 80% chance of detecting a real effect if it exists.

Power, Type I and Type II Errors

Metric	Symbol	Meaning
Type I Error	α	Rejecting a true H_0 (False Positive)
Type II Error	β	Failing to reject a false H_0 (False Negative)
Power	$1 - \beta$	Correctly rejecting a false H_0 (Detecting a real effect)

Key Takeaways

- Power = $1 - \beta$.
- Higher power reduces the risk of missing a real effect.
- Power increases with larger effect size, larger sample size, and higher α.
- Commonly used power values: 80% or 90%.
- Balance power with practical constraints (cost, time, resources).

Numerical Example with Excel Calculation

Example: A/B Test for Website Conversion Rate

A company wants to detect an increase in conversion rate from 5% to 7% (2% absolute increase). They plan to use a significance level of 0.05 and want to achieve 80% power. Let's calculate the required sample size using Excel.

Parameters	Value
Baseline conversion rate (p_1)	0.05
New conversion rate (p_2)	0.07
Effect size (Δ)	0.02
Significance level (α)	0.05
Desired power ($1 - \beta$)	0.80

Step 1: Calculate sample size (approx.)

$$n = \frac{(Z_{\alpha/2} + Z_{\beta})^2 \times \bar{p}(1 - \bar{p})}{\Delta^2}$$

where $\bar{p} = (p_1 + p_2)/2$

Excel Calculation:

$Z_{\alpha/2}$ (for 0.05)	=NORM.S.INV(1-0.05/2)	→ 1.96
Z_{β} (for 80% power)	=NORM.S.INV(0.80)	→ 0.84
$\bar{p}$	= (0.05 + 0.07)/2	→ 0.06
Δ	= 0.07 − 0.05	→ 0.02

$$n = \frac{(1.96 + 0.84)^2 \times 0.06 \times (1 - 0.06)}{0.02^2}$$

$n \approx 456$ per group

Step 2: Interpret the Result

- We need approximately 456 users in each group to achieve 80% power to detect a 2% increase in conversion rate at 5% significance level.
- If we use a smaller sample size, the power will be lower, and we may fail to detect the real improvement.

Excel Formula (One-line)

```
=((NORM.S.INV(1-0.05/2)+NORM.S.INV(0.80))^2*0.06*(1-0.06))/0.02^2
```

Result: 456 (per group)

Business Interpretation

To confidently detect a 2% increase in conversion rate, we should run the test with at least 456 users in each group. This ensures 80% power to find a real improvement.

13 Statistical Significance vs Practical Significance

Is the Result Really Meaningful in Business?

What is Statistical Significance?

A result is statistically significant if the p-value is less than the significance level (α). It means we have enough evidence to reject the null hypothesis (H_0).

> "Statistical significance tells us that an effect exists, but it does not tell us how big or important the effect is."

What is Practical Significance?

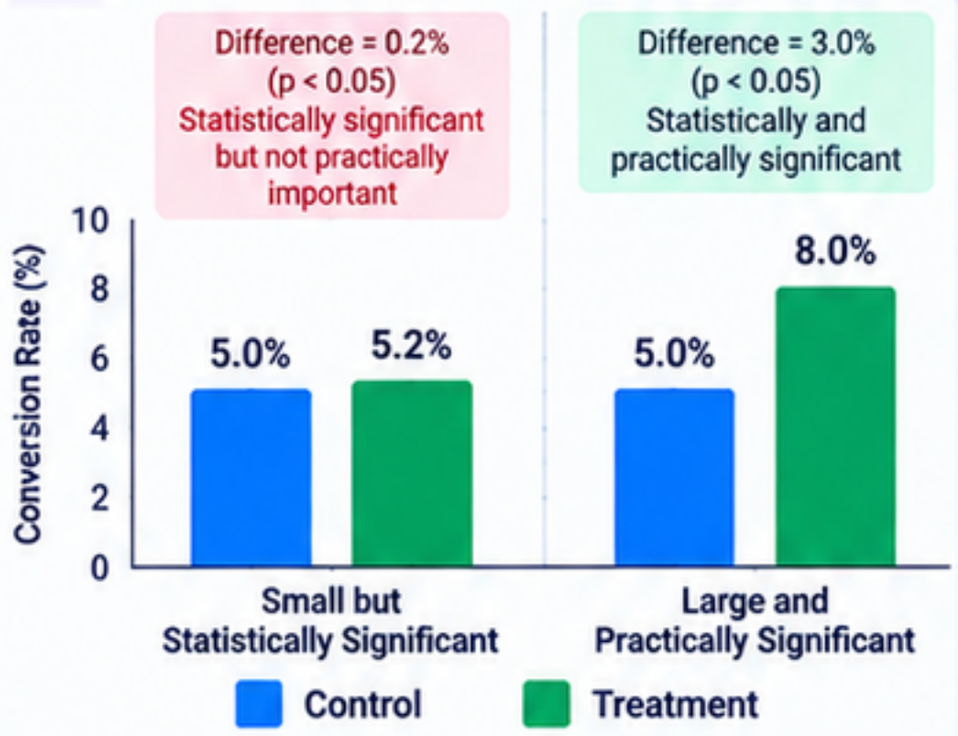
A result is practically significanct if the effect size is large enough to be meaningful and valuable in a real business or real-world context.

> "Practical significance considers whether the size of the effect is actually important in practice."

Why is this Important?

- A result can be statistically significant but not practically significant.
- Helps avoid taking actions on trivial effects.
- Important for business decision-making and resource allocation.
- Considers both statistical results and real-world impact (cost, benefit, feasibility).
- Common in A/B testing, product improvements, and policy analysis.

Visual Example: Small vs Large Effect



Comparison Table

Aspect	Statistical Significance	Practical Significance
Definition	Result unlikely due to chance (p-value < α)	Effect size is large enough to be meaningful
Focus	Evidence of any effect	Magnitude and real-world impact
Key Metric	p-value	Effect size (absolute or relative difference)
Answer to	Is there an effect?	Is the effect important?
Example	p < 0.05	Increase in sales by 5% or more

Effect Size for Practical Significance

Effect size measures the magnitude of the difference, not just whether it exists.

Type of Data	Effect Size Measure	Interpretation
Continuous data	Cohen's d	0.2 = small 0.5 = medium 0.8 = large
Proportions (e.g., conversion rate)	Difference in proportions (Δp)	Check if the increase is meaningful for the business (e.g., 1% vs 5%).
Categorical data	Cramér's V (or Odds Ratio)	Measure the strength of association.
General rule	Combine p-value and effect size	A result should be both statistically and practically significant.

Real-World Business Examples

Business Area	Scenario	Result	Interpretation
E-commerce	New website design increases conversion rate from 5.0% to 5.2%	p = 0.03 (Δ = 0.2%)	Statistically significant but not practically significant (small impact).
Marketing	New ad campaign increases sales from ₹10L to ₹13L	p = 0.01 (Δ = 30%)	Statistically and practically significant (large impact).
Manufacturing	Reduces defect rate from 2.0% to 1.9%	p = 0.04 (Δ = 0.1%)	Statistically significant but may not be practically important (tiny improvement).
HR Analytics	New training program increases productivity score from 70 to 71	p = 0.02 (Δ = 1 point)	Statistically significant but not practically important.
Finance	New fraud detection reduces fraud cases from 5% to 3%	p = 0.01 (Δ = 2%)	Statistically and practically significant (meaningful cost savings).

Numerical Example with Excel

A company tests a new ad campaign. Control group conversion rate = 5.0%, Treatment group = 5.2%. Sample size in each group = 5,000, α = 0.05.

Parameter	Value
Control conversion rate (p_1)	0.05
Treatment conversion rate (p_2)	0.052
Difference (Δp)	0.002 (0.2%)
Sample size ($n_1 = n_2$)	5,000
Significance level (α)	0.05

Calculation (Two-proportion Z-test)

$$\bar{p} = \frac{p_1 + p_2}{2} = \frac{0.05 + 0.052}{2} = 0.051$$

$$SE = \sqrt{\bar{p}(1-\bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$= \sqrt{0.051 \times 0.949 \times \left(\frac{1}{5000} + \frac{1}{3000}\right)} = 0.00440$$

$$z = \frac{p_2 - p_1}{SE} = \frac{0.002}{0.00440} = 0.45$$

p-value = 0.653 (using Excel: =2*(1-NORM.S.DIST(0.45,TRUE)))

Result: p = 0.653 > 0.05 → Not statistically significant and not practically significant.

When to Consider Practical Significance?

- When the sample size is very large (small differences can become statistically significant).
- When making business decisions with cost and resource implications.
- When evaluating real-world impact (e.g., revenue, customer satisfaction, productivity).
- In A/B testing, marketing campaigns, product changes, and policy analysis.

Key Takeaways

- Statistical significance (p-value) tells us if an effect exists.
- Practical significance tells us if the effect is meaningful.
- A result should ideally be both statistically and practically significant.
- Always consider effect size and business context.
- Large sample size can make small, unimportant differences statistically significant.
- Make data-driven decisions, not just statistically-driven decisions.

14 Steps in Hypothesis Testing

A Complete Step-by-Step Process with Example

What are the Steps in Hypothesis Testing?

Hypothesis testing follows a systematic step-by-step process to make a decision about the null hypothesis (H_0) using sample data and statistical methods.

> "It helps us go from a business question to a clear, data-driven conclusion."

Key Points

- Follow a structured process from start to finish.
- Start with a business problem and define hypotheses.
- Choose the right significance level (α).
- Select the appropriate test and calculate test statistic.
- Make a decision and interpret the result in a real business context.

Why is it Important?

- Ensures a logical and systematic approach.
- Reduces chances of mistakes and wrong conclusions.
- Helps in clear communication of results.
- Widely used in business, research, analytics, and decision-making.
- Makes the testing process repeatable and reliable.

1 State the Problem

- Clearly define the business question or problem.
- Identify the parameter of interest (mean, proportion, etc.).

Example: Has the average delivery time reduced after process improvement?

2 Set up the Hypotheses

- State the null hypothesis (H_0) and alternative hypothesis (H_1).
- H_0 usually represents no effect or no difference.
- H_1 represents the claim or effect we want to test.

Example:
H_0: $\mu = 5$ days
H_1: $\mu < 5$ days (delivery time reduced)

3 Choose Significance Level (α)

- Decide the level of significance (α), commonly 0.05.
- It is the maximum risk of rejecting a true H_0 (Type I error).

Example:
$\alpha = 0.05$ (5% risk of Type I error)

4 Choose the Appropriate Test

- Select the correct statistical test based on:
  - Type of data (mean, proportion, categorical, etc.)
  - Sample size (small or large)
  - Distribution (normal or not)
  - Number of groups (one or two)
  - Type of test (one-tailed or two-tailed)

Example: One-sample t-test (for mean, small sample size)

5 Collect and Analyze the Data

- Collect sample data.
- Calculate the test statistic using the appropriate formula.
- Find the p-value or compare with critical value.

Example:
Sample mean ($\bar{x}$) = 4.7 days
n = 25, s = 1.2 days

6 Make a Decision

- If p-value $\leq \alpha \rightarrow$ Reject H_0 (statistically significant).
- If p-value $> \alpha \rightarrow$ Fail to Reject H_0 (not statistically significant).

Example:
p-value = 0.021 $\leq$ 0.05
$\rightarrow$ Reject H_0

7 Interpret the Result

- Explain what the decision means in the business context.
- Mention whether the result is statistically significant.
- Discuss the practical significance (effect size, impact).

Example:
There is sufficient evidence that the average delivery time has reduced, which can improve customer satisfaction.

8 Take Action

- Based on the result, take an appropriate business action.
- If significant $\rightarrow$ Implement the change.
- If not significant $\rightarrow$ Re-evaluate the strategy or collect more data.

Example:
Since delivery time is reduced, the company continues the new process and scales it across other regions.

Flowchart: Steps in Hypothesis Testing

1. Define the Problem (Business Question)
2. Set up Hypotheses (H_0 and H_1)
3. Choose Significance Level (α)
4. Select Appropriate Test
5. Collect and Analyze Data (Calculate Test Statistic & p-value)
6. Make a Decision (Reject or Fail to Reject H_0)
7. Interpret the Result (Business Context)
8. Take Action (Implement / Re-evaluate)

Numerical Example with Excel Calculation

A company wants to check if the average delivery time has reduced from 5 days to less than 5 days after a new logistics process. Sample of 25 deliveries has a mean of 4.7 days and standard deviation of 1.2 days. Use $\alpha = 0.05$.

1 Input Data

Parameter	Value
Hypothesized Mean (μ_0)	5
Sample Mean ($\bar{x}$)	4.7
Sample Std Dev (s)	1.2
Sample Size (n)	25
Significance Level (α)	0.05
Type of Test	Left-tailed (t-test)

2 Excel Calculation

Test statistic (t):

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{4.7 - 5}{1.2/\sqrt{25}} = \frac{-0.3}{0.24} = -1.25$$

p-value (left-tailed):
=T.DIST(−1.25, 24, TRUE)

p-value = 0.111

3 Decision and Interpretation

p-value (0.111) > 0.05 $\rightarrow$ Fail to Reject H_0
There is not enough evidence to conclude that the average delivery time has reduced. The result is not statistically significant at 5% level.

Key Takeaways

- Follow a structured step-by-step process.
- Clearly define the problem and hypotheses.
- Choose the correct significance level and test.
- Use sample data to calculate the test statistic and p-value.
- Make a decision based on p-value or critical value.
- Always interpret the result in real business context.
- Take appropriate action based on the findings.

Common Mistakes to Avoid

- Choosing the wrong test.
- Misinterpreting the p-value.
- Ignoring practical significance.
- Not clearly stating the hypotheses.
- Drawing conclusions without business context.

15 Business Interpretation of Hypothesis Testing

From Statistical Results to Real-World Decisions

What is Business Interpretation?

Business interpretation means converting the statistical results (p-value, test statistic, etc.) into meaningful business conclusions and actions.

> "Statistics gives us the evidence, but business interpretation helps us decide what to do with that evidence."

Key Points

- Look beyond just p-value.
- Consider effect size and practical significance.
- Evaluate the business context and real-world impact.
- Assess cost, risk, and feasibility of action.
- Make data-driven decisions (continue, change, or re-evaluate).
- Clearly communicate the findings to stakeholders.

Why is it Important?

- Helps in making practical and profitable decisions.
- Avoids overreacting to statistically significant but trivial results.
- Considers real business constraints (cost, time, resources).
- Ensures data-driven and objective decision-making.
- Builds trust with stakeholders through clear communication.
- Widely used in business, analytics, marketing, finance, and operations.

From Statistical Result to Business Meaning

Statistical Result	Interpretation Step	Business Decision
• p-value • Test statistic • Confidence interval • Effect size • Decision (Reject / Fail to Reject H_0)	• Is the result statistically significant? • What is the magnitude of the effect? • Is the effect practically important? • What are the business implications?	• Continue current strategy • Implement change • Re-evaluate with more data • Consider cost, risk, and feasibility

Common Scenarios and Interpretation

Scenario	Statistical Result	Business Interpretation	Possible Action
Statistically Significant & Practically Significant	p-value < 0.05 and large effect size	Real and meaningful difference exists.	Implement the change. Scale the strategy.
Statistically Significant but Not Practically Significant	p-value < 0.05 but small effect size	Difference exists but is too small to be meaningful.	No major change. May monitor or collect more data.
Not Statistically Significant (Null Not Rejected)	p-value ≥ 0.05	No sufficient evidence of a real difference.	Continue current strategy or collect more data.
Type II Error Risk (Low Power)	p-value ≥ 0.05 with small sample size	Effect may exist but was not detected.	Increase sample size and re-test.

Real-World Example 1: E-commerce (A/B Test)

A company tests a new website design to increase conversion rate.

- H_0: No difference in conversion rate.
- H_1: New design increases conversion rate.
- **Result:** p-value = 0.03, increase = 0.8% (from 2.0% to 2.8%).
- **Interpretation:** Statistically significant and practically significant (0.8% increase is meaningful for revenue).
- **Business Action:** Implement new design and roll it out to all users.

Real-World Example 2: Marketing Campaign

A company runs a new ad campaign to increase sales.

- H_0: No difference in sales.
- H_1: New campaign increases sales.
- **Result:** p-value = 0.04, increase = 1% (from ₹100 to ₹101).
- **Interpretation:** Statistically significant but practically not significant (1% increase is too small compared to campaign cost ₹5,00,000).
- **Business Action:** Do not implement immediately. Re-evaluate campaign or test with a larger change.

Real-World Example 3: Operations (Delivery Time)

A logistics company tests a new process to reduce delivery time.

- H_0: No difference in average delivery time.
- H_1: New process reduces delivery time.
- **Result:** p-value = 0.12, reduction = 0.6 days (from 5.0 to 4.4 days).
- **Interpretation:** Not statistically significant (insufficient evidence to confirm a real reduction).
- **Business Action:** Continue current process or collect more data with a larger sample size.

Key Takeaways

- Statistical significance does not always mean practical significance.
- Always consider effect size and business context.
- Small statistically significant changes may not be worth the cost or effort.
- Use hypothesis testing as a tool for evidence-based decision-making.
- Clearly communicate results and recommendations to stakeholders.

Common Mistakes to Avoid

- Focusing only on p-value.
- Ignoring effect size and practical significance.
- Making decisions without considering cost, risk, and feasibility.
- Overreacting to small statistically significant results.
- Not considering the possibility of Type II error (low power).
- Misinterpreting non-significant results as proof of no effect.

Interview Questions

1. How do you interpret a statistically significant result in a business context?
2. What would you do if the result is statistically significant but not practically significant?
3. How do you explain a non-significant result to a non-technical stakeholder?
4. Why is business interpretation important in hypothesis testing?

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MODULE 8
HYPOTHESIS TESTING BASICS
From Assumption to Evidence

"Better Analysts Make a Smarter Tomorrow.."

Learn Apply Grow

Final Summary, Real-World Applications & Interview Questions

Key Takeaways, Practical Use Cases and Expert Q&A for Hypothesis Testing

What We Learned in This Module

- What is hypothesis testing and why it is important.
- Null and alternative hypothesis (H_0 and H_1).
- One-tailed and two-tailed tests.
- Type I and Type II errors (α and β).
- P-value approach and critical value approach.
- Decision rule and interpretation of results.
- Power of a test and factors affecting power.
- Statistical significance vs practical significance.
- Step-by-step process of hypothesis testing.
- Business interpretation with real-world examples.

Key Formulas to Remember

Concept	Formula
Test Statistic (Z)	$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$
Test Statistic (t)	$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$
P-value	P = P(Test statistic ≥ observed value) (or ≤ for left-tailed)
Critical Value (Z)	$Z_{\alpha/2}, Z_{\alpha}$ (from standard normal table or Excel)
Critical Value (t)	$t_{\alpha/2,df}, t_{\alpha,df}$ (from t table or Excel)
Power of a Test	Power = $1 - \beta$
Effect Size (Cohen's d)	$d = \frac{\mu_1 - \mu_2}{s_p}$

Steps in Hypothesis Testing (Recap)

1. State the problem and define the parameter.
2. Set up null and alternative hypotheses (H_0 & H_1).
3. Choose the significance level (α).
4. Select the appropriate statistical test.
5. Collect and analyze the data.
6. Calculate the test statistic and p-value (or compare with critical value).
7. Make a decision (Reject or Fail to Reject H_0).
8. Interpret the result in business context.
9. Take action and communicate the findings.

Real-World Applications

Retail & E-commerce	Test if a new website design increases conversion rate. Compare average order value between two campaigns.
Marketing	Evaluate if a new ad campaign improves sales. Test if customer engagement has increased.
Operations	Check if a process improvement reduces delivery time. Compare defect rates before and after a change.
Finance	Test if the average revenue has increased. Analyze if costs have reduced significantly.
HR & People Analytics	Test if a new training program improves employee performance. Compare attrition rates before and after a policy change.
Healthcare	Check if a new treatment is more effective than the existing one. Compare recovery rates between two groups.

Key Concepts at a Glance

Concept	Key Idea
Null Hypothesis (H_0)	No effect or no difference (baseline assumption).
Alternative Hypothesis (H_1)	There is an effect or a difference.
Significance Level (α)	Maximum risk of rejecting a true H_0 (commonly 0.05).
P-value	Probability of observing the result if H_0 is true.
Type I Error (α)	Rejecting a true H_0 (False Positive).
Type II Error (β)	Failing to reject a false H_0 (False Negative).
Power (1 – β)	Probability of correctly detecting a real effect.
Effect Size	Magnitude of the difference (e.g., Cohen's d).
Statistical Significance	Result is unlikely due to chance (p-value < α).
Practical Significance	Result is meaningful and valuable in real business context.

Common Business Scenarios

Scenario	Test Type	What We Check
Compare average sales before and after a campaign	One-sample t-test	Is there a significant increase in average sales?
Compare two product conversion rates	Two-sample Z-test (proportions)	Is the new product conversion rate higher?
Check if delivery time has reduced	Two-sample t-test	Is the average delivery time lower after improvement?
Compare defect rates in two batches	Two-proportion Z-test	Is the defect rate significantly lower?
Evaluate training program effectiveness	Paired t-test	Is there a significant improvement in scores?
Test if cost reduction is real	One-sample t-test	Is the average cost significantly lower than the target?
A/B testing for website design	Two-proportion Z-test	Is the new design performing better in conversion rate?

Decision Flowchart

Key Takeaways

- Hypothesis testing helps us make data-driven decisions.
- A statistically significant result does not always mean practical significance.
- Always consider effect size, business context, and cost-benefit.
- Larger sample size increases the power of a test.
- Control Type I and Type II errors based on business risk.
- Use the correct test based on the data type and scenario.
- Always validate assumptions (normality, independence, etc.).
- Clearly communicate the results with business implications.
- Hypothesis testing is widely used in analytics, research, and decision-making.

"Statistics turns data into evidence, and evidence into better decisions."
- Tech With Shivansh

Interview Questions with Answers

Q: What is hypothesis testing?
A: A statistical method used to make a decision about a population parameter based on sample data.

Q: What is the null hypothesis?
A: It states that there is no effect or no difference.

Q: What is the significance level?
A: The maximum risk of rejecting a true null hypothesis, commonly 0.05 (5%).

Q: What is a p-value?
A: The probability of observing the result if the null hypothesis is true.

Q: What is Type I error?
A: Rejecting a true null hypothesis (false positive).

Q: What is Type II error?
A: Failing to reject a false null hypothesis (false negative).

Q: What is the power of a test?
A: The probability of correctly rejecting a false null hypothesis ($1 - \beta$).

Q: How do you choose the right test?
A: Based on the type of data, sample size, distribution, number of groups, and the research question.